

UNIVERSITY OF KARACHI

Department of Computer Science (UBIT)

PROJECT REPORT

Predictive Analysis System for Karachi Air Quality

Machine Learning-Based Air Quality Forecasting

Course Information

Course Title: Artificial Intelligence
Course Code: CS-551
Section: A
Department: UBIT (Computer Science)

Project Team Members

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1 Introduction

Air pollution is a major environmental and health concern, especially in urban cities like Karachi. Fine particulate matter (PM_{2.5}) is one of the most harmful pollutants, as it can penetrate deep into the lungs and bloodstream, causing serious health issues such as asthma, bronchitis, lung cancer, and cardiovascular diseases, which can lead to reduced lung function, breathing difficulties, heart attacks, and even premature death. Long-term exposure to high levels of PM_{2.5} significantly increases the risk of chronic respiratory and heart conditions, making it a critical public health issue.

This project aims to analyze air quality data collected from Aga Khan University, Karachi, and develop machine learning models to predict PM_{2.5} levels based on environmental and sensor data.

2 Problem Definition

The objective of this project is to:

- Analyze air quality data
- Identify factors affecting pollution levels
- Build machine learning models to predict PM_{2.5}
- Evaluate model performance
- Provide insights and improvements

3 Data Collection

The dataset was obtained from air quality sensors located at Aga Khan University, Karachi.

Dataset Features:

- PM₁ (fine particles)
- PM_{2.5} (target variable)
- Temperature
- Relative Humidity
- UM003 (ultra-fine particle count)
- Date and time

The dataset represents real-world environmental conditions and contains time-series data.

4 Data Preprocessing

The following preprocessing steps were performed:

- Converted raw dataset into structured format using pivot tables
- Handled missing values using mean imputation
- Converted datetime column into usable format
- Extracted time-based features (hour, day, month)
- Removed redundant features (UM003, month) based on correlation analysis

5 Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and relationships in the data.

Techniques used:

- Correlation matrix
- Heatmap visualization
- Time-series plots

Key Insights:

- PM_1 has a very strong positive correlation with $PM_{2.5}$ (~ 0.99)
- Temperature has a slight negative relationship with pollution
- Humidity has a moderate positive relationship with pollution
- Pollution levels vary over time

6 Feature Engineering

New features were created to improve model performance:

- Pollution Index = $PM_1 + PM_{2.5}$
- Temperature-Humidity Ratio = Temperature/Humidity

Feature engineering was performed to enhance model performance removing irrelevant or redundant features as well. However, care was taken to avoid data leakage, as $PM_{2.5}$ is the target variable. Therefore, features directly derived from $PM_{2.5}$ (such as Pollution Index) were not used in model training. Instead, only independent variables such as PM_1 , temperature, humidity, and time-based features were utilized.

7 Model Building

The dataset was split into training and testing sets to evaluate model performance on unseen data. An 80:20 ratio was used, where 80% of the data was used for training the models and 20% was reserved for testing. This ensures that the model generalizes well and avoids overfitting.

Three machine learning models were implemented:

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor

These models were trained using the processed dataset to predict $PM_{2.5}$ levels.

8 Model Evaluation

Models were evaluated using:

- R^2 Score (accuracy measure)
- Mean Squared Error (error measure)

Results:

- Random Forest achieved the best performance

9 Conclusion

This project successfully demonstrated the use of machine learning techniques to predict air pollution levels.

- PM_1 was identified as the strongest predictor of $PM_{2.5}$
- Environmental factors like temperature and humidity also influenced pollution
- Random Forest provided the highest prediction accuracy due to its ability to model complex relationships

10 Limitations

- Dataset is limited to a single location (Aga Khan University)
- Lack of additional environmental data (wind speed, rainfall)
- Highly correlated features may affect model generalization

11 Future Improvements

- Incorporate additional weather data (wind speed, rainfall)
- Collect data from multiple locations across Karachi
- Perform hyperparameter tuning for better accuracy

12 User Interface

A simple interactive interface was developed using Python widgets, allowing users to input environmental parameters such as PM_{10} , temperature, humidity, hour, and day.

The system predicts $PM_{2.5}$ levels in real-time, making the model user-friendly and practical for demonstration purposes.

13 Tools & Technologies

- Python
- Google Colab
- Pandas
- Matplotlib, Seaborn
- Scikit-learn
- ipywidgets (for interface)

Project Links

- **GitHub Repository:**
<https://github.com/HusainAhmadKhan/KarachiAirPollutionPrediction>
- **Google Colab Notebook:**
<https://colab.research.google.com/drive/1p0aH5ySZGrLICUe4aH5UD74VSPLCIS74?usp=sharing>